

cup product structures we can conclude that the Pontryagin rings $H_*(\mathbb{R}P^\infty; \mathbb{Z}_2)$ and $H_*(\mathbb{C}P^\infty; \mathbb{Z})$ are divided polynomial algebras.

In these examples the Hopf algebra is generated as an algebra by primitive elements, so the product determines the coproduct and hence the dual algebra. This is not true in general, however. For example, we have seen that the Hopf algebra $\Gamma_{\mathbb{Z}_p}[\alpha]$ is isomorphic as an algebra to $\bigotimes_{i \geq 0} \mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$, but if we regard the latter tensor product as the tensor product of the Hopf algebras $\mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$ then the elements α_{p^i} are primitive, though they are not primitive in $\Gamma_{\mathbb{Z}_p}[\alpha]$ for $i > 0$. In fact, the Hopf algebra $\bigotimes_{i \geq 0} \mathbb{Z}_p[\alpha_{p^i}]/(\alpha_{p^i}^p)$ is its own dual, according to one of the exercises below, but the dual of $\Gamma_{\mathbb{Z}_p}[\alpha]$ is $\mathbb{Z}_p[\alpha]$.

Exercises

1. Suppose that X is a CW complex with basepoint $e \in X$ a 0-cell. Show that X is an H-space if there is a map $\mu: X \times X \rightarrow X$ such that the maps $X \rightarrow X$, $x \mapsto \mu(x, e)$ and $x \mapsto \mu(e, x)$, are homotopic to the identity. [Sometimes this is taken as the definition of an H-space, rather than the more restrictive condition in the definition we have given.] With the same hypotheses, show also that μ can be homotoped so that e is a strict two-sided identity.
2. Show that a retract of an H-space is an H-space if it contains the identity element.
3. Show that in a homotopy-associative H-space whose set of path-components is a group with respect to the multiplication induced by the H-space structure, all the path-components must be homotopy equivalent. [Homotopy-associative means associative up to homotopy.]
4. Show that an H-space or topological group structure on a path-connected, locally path-connected space can be lifted to such a structure on its universal cover. [For the group $SO(n)$ considered in the next section, the universal cover for $n > 2$ is a 2-sheeted cover, a group called $Spin(n)$.]
5. Show that if (X, e) is an H-space then $\pi_1(X, e)$ is abelian. [Compare the usual composition $f \cdot g$ of loops with the product $\mu(f(t), g(t))$ coming from the H-space multiplication μ .]
6. Show that S^n is an H-space iff the attaching map of the $2n$ -cell of $J_2(S^n)$ is homotopically trivial.
7. What are the primitive elements of the Hopf algebra $\mathbb{Z}_p[x]$ for p prime?
8. Show that the tensor product of two Hopf algebras is a Hopf algebra.
9. Apply the theorems of Hopf and Borel to show that for an H-space X that is a connected finite CW complex with $\tilde{H}_*(X; \mathbb{Z}) \neq 0$, the Euler characteristic $\chi(X)$ is 0.
10. Let X be a path-connected H-space with $H^*(X; R)$ free and finitely generated in each dimension. For maps $f, g: X \rightarrow X$, the product $fg: X \rightarrow X$ is defined by $(fg)(x) = f(x)g(x)$, using the H-space product.

- (a) Show that $(fg)^*(\alpha) = f^*(\alpha) + g^*(\alpha)$ for primitive elements $\alpha \in H^*(X; R)$.
- (b) Deduce that the k^{th} -power map $x \mapsto x^k$ induces the map $\alpha \mapsto k\alpha$ on primitive elements α . In particular the quaternionic k^{th} -power map $S^3 \rightarrow S^3$ has degree k .
- (c) Show that every polynomial $a_n x^n b_n + \cdots + a_1 x b_1 + a_0$ of nonzero degree with coefficients in $\mathbb{H}$ has a root in $\mathbb{H}$. [See Theorem 1.8.]
11. If T^n is the n -dimensional torus, the product of n circles, show that the Pontryagin ring $H_*(T^n; \mathbb{Z})$ is the exterior algebra $\Lambda_{\mathbb{Z}}[x_1, \dots, x_n]$ with $|x_i| = 1$.
12. Compute the Pontryagin product structure in $H_*(L; \mathbb{Z}_p)$ where L is an infinite-dimensional lens space $S^\infty/\mathbb{Z}_p$, for p an odd prime, using the coproduct in $H^*(L; \mathbb{Z}_p)$.
13. Verify that the Hopf algebras $\Lambda_R[\alpha]$ and $\mathbb{Z}_p[\alpha]/(\alpha^p)$ are self-dual.
14. Show that the coproduct in the Hopf algebra $H_*(X; R)$ dual to $H^*(X; R)$ is induced by the diagonal map $X \rightarrow X \times X$, $x \mapsto (x, x)$.
15. Suppose that X is a path-connected H-space such that $H^*(X; \mathbb{Z})$ is free and finitely generated in each dimension, and $H^*(X; \mathbb{Q})$ is a polynomial ring $\mathbb{Q}[\alpha]$. Show that the Pontryagin ring $H_*(X; \mathbb{Z})$ is commutative and associative, with a structure uniquely determined by the ring $H^*(X; \mathbb{Z})$.
16. Classify algebraically the Hopf algebras A over $\mathbb{Z}$ such that A^n is free for each n and $A \otimes \mathbb{Q} \approx \mathbb{Q}[\alpha]$. In particular, determine which Hopf algebras $A \otimes \mathbb{Z}_p$ arise from such A 's.

3.D The Cohomology of $SO(n)$

After the general discussion of homological and cohomological properties of H-spaces in the preceding section, we turn now to a family of quite interesting and subtle examples, the orthogonal groups $O(n)$. We will compute their homology and cohomology by constructing very nice CW structures on them, and the results illustrate the general structure theorems of the last section quite well. After dealing with the orthogonal groups we then describe the straightforward generalization to Stiefel manifolds, which are also fairly basic objects in algebraic and geometric topology.

The orthogonal group $O(n)$ can be defined as the group of isometries of $\mathbb{R}^n$ fixing the origin. Equivalently, this is the group of $n \times n$ matrices A with entries in $\mathbb{R}$ such that $AA^t = I$, where A^t is the transpose of A . From this viewpoint, $O(n)$ is topologized as a subspace of $\mathbb{R}^{n^2}$, with coordinates the n^2 entries of an $n \times n$ matrix. Since the columns of a matrix in $O(n)$ are unit vectors, $O(n)$ can also be regarded as a subspace of the product of n copies of S^{n-1} . It is a closed subspace since the conditions that columns be orthogonal are defined by polynomial equations. Hence